

Dynamics of Neuronal Gamma Oscillations on a Random Graph

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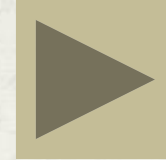
Oct 13, 2023

Various Synchronization

- Firefly flashing
- Neural network
- Cardiac muscle cell
- Network of power plants
- Locomotion



Firefly



Metronome



Pendulum



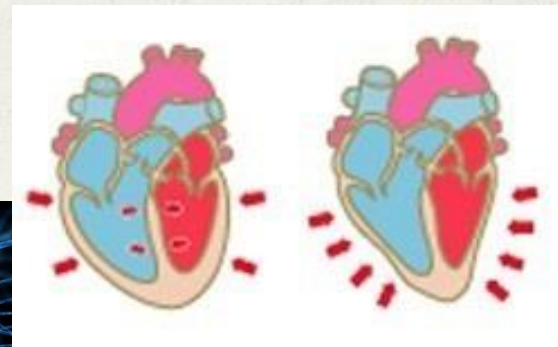
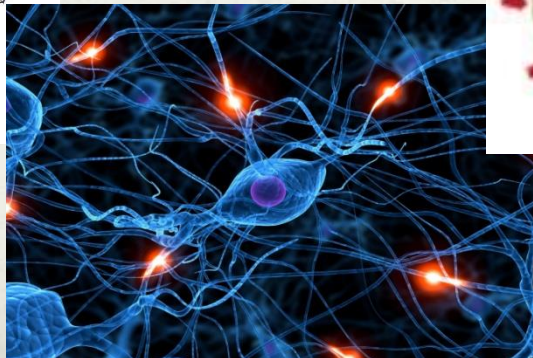
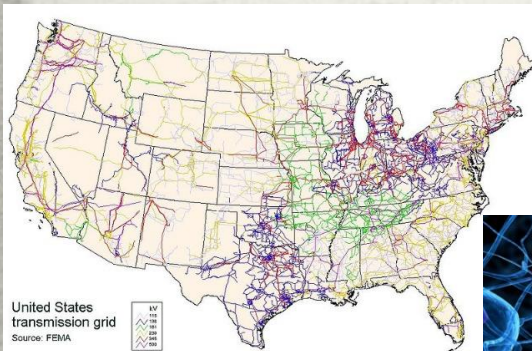
Bottoles



Bridge



Cat



Kuramoto model (1975)

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N.$$

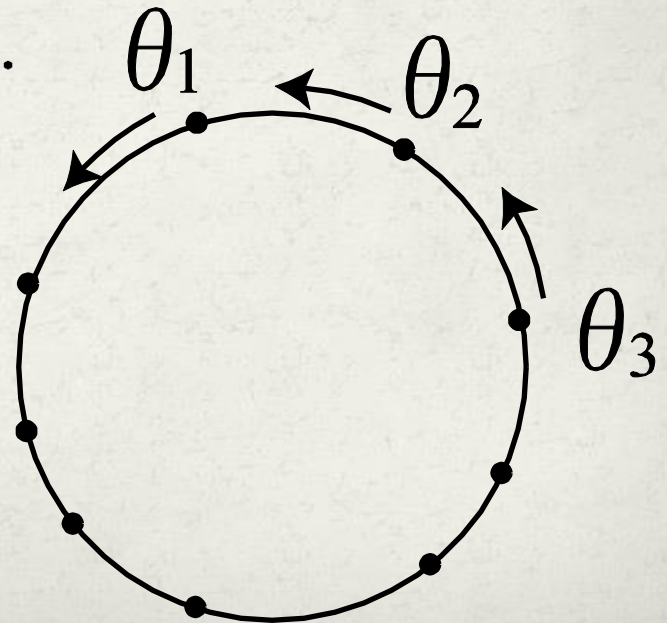
$$\theta_i \in S^1$$

N : the number of particles.

ω_i : natural frequency of θ_i .

$K > 0$: coupling strength.

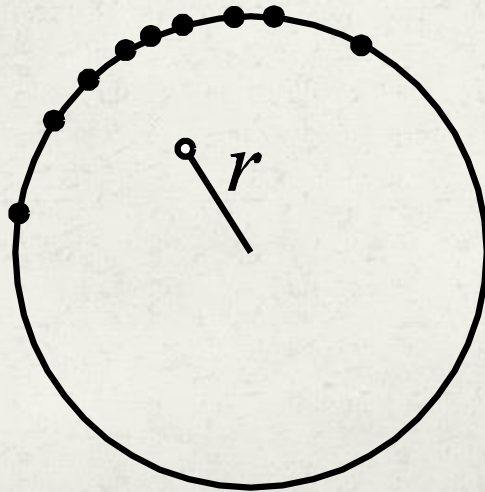
If $K > K_c$,
synchronization occurs.



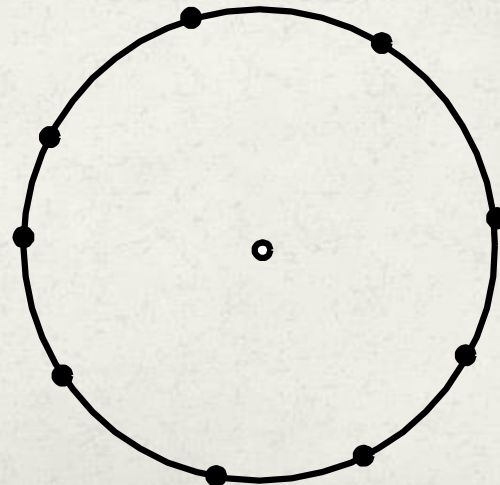
Kuramoto model (1975)

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N.$$

Order parameter : $re^{i\psi} := \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)}$



synchro



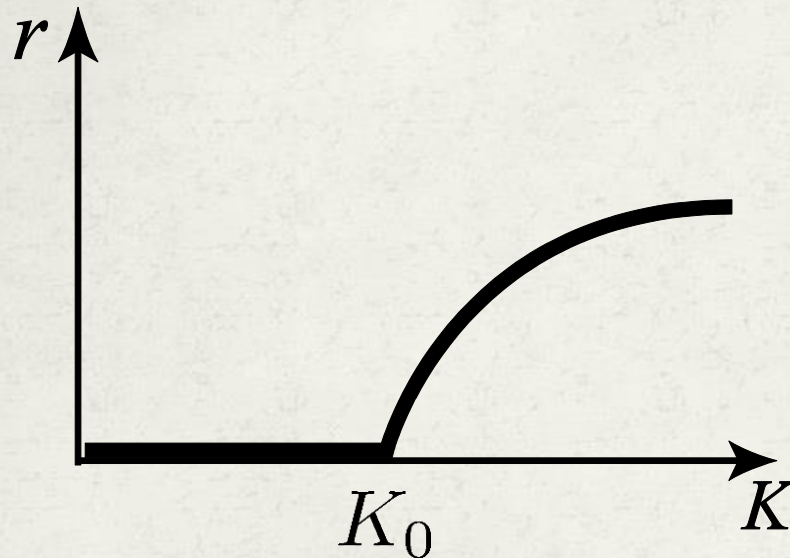
de-synchro

Kuramoto conjecture (1984)

$$N \rightarrow \infty,$$

$g(\omega)$: distribution function for ω_i 's.

If $g(\omega)$ is even and unimodal, then



$$K_0 := \frac{2}{\pi g(0)}$$

The generalized spectral theory is developed to solve the conjecture.

H.Chiba,

A proof of the Kuramoto conjecture of the infinite dimensional Kuramoto model,
Ergo. Theo. Dyn. Syst, (2015)

A spectral theory of linear operators on rigged Hilbert spaces,
Adv. in Math, (2015)

Purpose

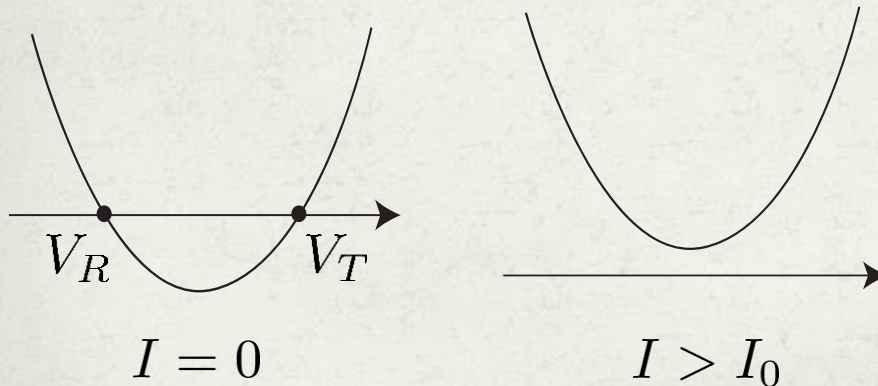
Application to neural networks.

- Existence/stability of gamma oscillations, which is related to cognition and attention.
- Local area of the brain cortex ($N \sim O(1000)$).
- Only inhibitory neurons.
(in progress: with excitatory neurons)
- Network structure is the Erdos-Reiny random with the connection probability p (bifurcation parameter).

QIF model (dynamics of a single inhibitory neuron)

$$\frac{dV}{dt} = g_L \frac{(V - V_R)(V - V_T)}{V_T - V_R} + I$$

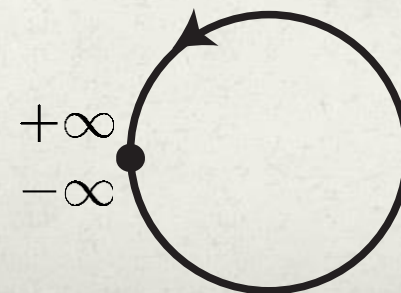
$V(t)$	membrane potential	Unknown func.
V_R	resting potential	-62 (mV)
V_T	threshold potential	-55 (mV)
g_L	leak conductance	0.1 (mS/cm ²)
I	Input current	drawn from a distribution func. $G(I)$



If $I > I_0$, a solution goes to $+\infty$ at a finite time (firing), and back to $-\infty$.

Riccati equation is a unique holomorphic vector field on CP^1 .

➡ Theta model.



Modified theta model (Kotani et al. 2014)

$$\frac{dV_i}{dt} = g_L \frac{(V_i - V_R)(V_i - V_T)}{V_T - V_R} + I_i - g_{syn}^i(t)(V_i - V_{syn})$$

$$\frac{dg_{syn}^i}{dt} = -\frac{1}{\tau} g_{syn}^i + g_{peak} \sum_{conn} \sum_{fire} \delta(t - t_k)$$



$$V_i \mapsto \theta_i$$

$$\frac{d\theta_i}{dt} = -g_L \cos \theta_i + c_1(1 + \cos \theta_i)I_i + g_{syn}^i(t)(c_2(1 + \cos \theta_i) - \sin \theta_i)$$

V_R	resting potential	-62 (mV)
V_T	threshold potential	-55 (mV)
g_L	leak conductance	0.1 (mS/cm ²)
I_i	Input current	drawn from a distribution func. $G(I)$
$g_{syn}^i(t)$	synaptic conductance	Unkown function
V_{syn}	reversal potential	-70 (mV)
g_{peak}	peak conductance	0.0214 (mS/cm ²)
τ	decay time	5 (ms)

$c_1 = 2/7$
 $c_2 = -23/7$
 are constants
 determined by
 V_R, V_T, V_{syn} .

Continuous limit & Averaging

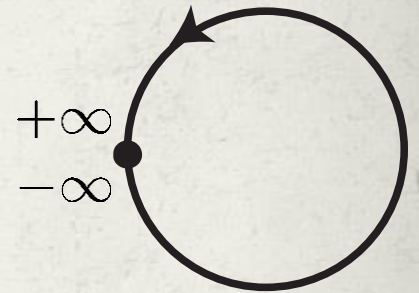
$$\frac{d\theta_i}{dt} = -g_L \cos \theta_i + c_1(1 + \cos \theta_i)I_i + g_{syn}^i(t)(c_2(1 + \cos \theta_i) - \sin \theta_i)$$

$$\frac{dg_{syn}^i}{dt} = -\frac{1}{\tau}g_{syn}^i + g_{peak} \sum_{conn} \sum_{fire} \delta(t - t_k)$$



Dynamics of the density function (Vlasov eq.)

$$\begin{cases} \frac{\partial P}{\partial t} + \frac{\partial}{\partial \theta}(vP) = 0, & P = P(t, \theta, I) \\ v(t, \theta, I) = -g_L \cos \theta + c_1(1 + \cos \theta)I + g_{syn}(t)(c_2(1 + \cos \theta) - \sin \theta) \end{cases}$$



Averaging on a random graph.

$$\frac{dg_s}{dt} = -\frac{1}{\tau}g_s + g_{peak} \cdot p \cdot N \int_{\mathbf{R}} v(t, \pi, I)P(t, \pi, I)G(I)dI$$

p : probability of connection. N : the number of neurons.

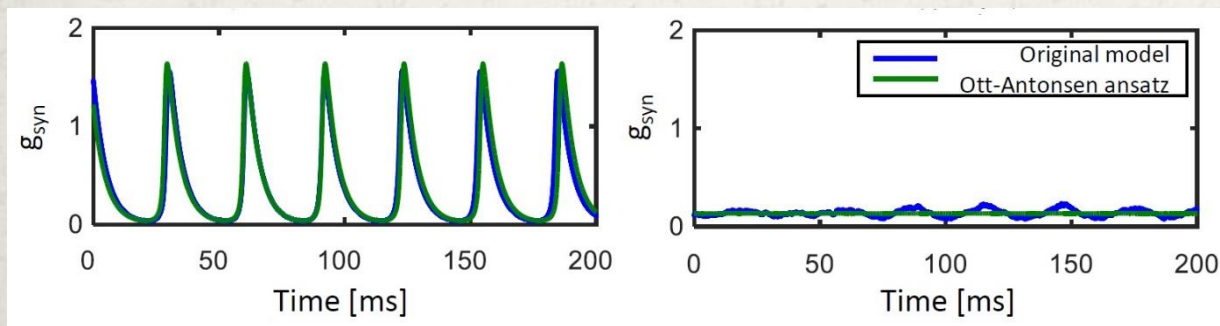
$G(I)$: distribution for current I .

Purpose.

$$\begin{cases} \frac{\partial P}{\partial t} + \frac{\partial}{\partial \theta}(vP) = 0, \quad P = P(t, \theta, I) \\ v(t, \theta, I) = -g_L \cos \theta + c_1(1 + \cos \theta)I + g_s(t)(c_2(1 + \cos \theta) - \sin \theta) \\ \frac{dg_s}{dt} = -\frac{1}{\tau}g_s + \mu \int_{\mathbf{R}} v(t, \pi, I)P(t, \pi, I)G(I)dI \end{cases}$$

$\mu = g_{peak} \cdot p \cdot N$: Complexity of the graph (bifurcation parameter).

A bifurcation from de-synchro (steady state) to the synchro (collective firing called the Gamma oscillation) as μ increases.



In what follows, suppose that $G(I)$ is symmetric and unimodal around the average $\eta > 0$.

Reduction by Ott-Antonsen ansatz.

$$\begin{cases} \frac{\partial P}{\partial t} + \frac{\partial}{\partial \theta}(vP) = 0, & P = P(t, \theta, I) \\ \frac{dg_s}{dt} = -\frac{1}{\tau}g_s + \mu \int_{\mathbf{R}} v(t, \pi, I)P(t, \pi, I)G(I)dI \end{cases}$$

Expand a solution in a Fourier series: $P = \frac{1}{2\pi} \sum z_k(t, I)e^{ik\theta}$

Then, $z_k = z_1^k$ satisfies the system and it is reduced to

$$\begin{cases} \dot{z}_1 = i(f(I)z_1^2 + h(I)z_1 + \overline{f(I)}) \\ \dot{\bar{z}}_1 = -i(\overline{f(I)} \bar{z}_1^2 + h(I)\bar{z}_1 + f(I)) \\ \dot{g}_s = -\frac{1}{\tau}g_s + \mu A(t). \end{cases}$$

Dynamical system on

$$\mathcal{H} = L^2(\mathbf{R}, G(I)dI) \oplus L^2(\mathbf{R}, G(I)dI) \oplus \mathbf{R}$$

where

$$\begin{cases} f(I) = -\frac{1}{2}g_L + \frac{1}{2}c_1I + \frac{1}{2}c_2g_s(t) + \frac{i}{2}g_s(t) \\ h(I) = c_1I + c_2g_s(t) \\ A(t) = \frac{g_L}{2\pi} \int \left(1 - \frac{z_1}{1+z_1} - \frac{\bar{z}_1}{1+\bar{z}_1} \right) G(I)dI \end{cases}$$

Steady state.

$$\begin{cases} \dot{z}_1 = i(f(I)z_1^2 + h(I)z_1 + \overline{f(I)}) \\ \dot{\bar{z}}_1 = -i(\overline{f(I)} \bar{z}_1^2 + h(I)\bar{z}_1 + f(I)) \\ \dot{g}_s = -\frac{1}{\tau}g_s + \mu A(t). \end{cases}$$

Steady state $(z_1, \bar{z}_1, g_s) = (z_0, \bar{z}_0, g_0)$ is unique, which corresponds to the de-synchro state.

We cannot write down it explicitly; g_0 is given as a root of

$$g_0 = \frac{\tau\mu}{4\pi g_L c_1} \int_{-\infty}^{\infty} x^2 G\left(\frac{1}{2g_L c_1}(x^2 + g_0^2 + g_L^2 - 2c_2 g_L g_0)\right) dx$$

The linearized system around the steady state:

$$\frac{d}{dt} \begin{pmatrix} u \\ \bar{u} \\ v \end{pmatrix} = \begin{pmatrix} Q(I) & 0 & \frac{P(I)}{P(I)} \\ 0 & \frac{Q(I)}{R(I)} & \frac{P(I)}{P(I)} \\ R(I) & \frac{Q(I)}{R(I)} & -1/\tau \end{pmatrix} \begin{pmatrix} u \\ \bar{u} \\ v \end{pmatrix} := T(\mu) \begin{pmatrix} u \\ \bar{u} \\ v \end{pmatrix}$$

Stability of the steady state  spectrum of the operator T .

Spectrum of the operator T .

The continuous spectrum: $\sigma_c(T) = i\mathbf{R} \cup \mathbf{R}_{\leq 0}$

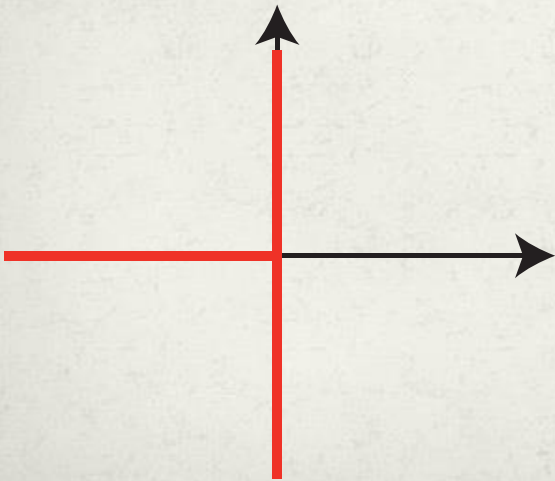
Eigenvalues λ are given as roots of the equation (if it exists)

$$\lambda + \frac{1}{\tau} = \frac{\mu}{4\pi g_L c_1} (\lambda + g_0 - c_2 g_L) \int \frac{i\omega}{\lambda - i\omega} \tilde{G}(\omega) d\omega$$

where

$$\tilde{G}(\omega) = G \left(\frac{1}{2g_L c_1} (\omega^2 + g_0^2 + g_L^2 - 2c_2 g_L g_0) \right)$$

We can show that there are no eigenvalues if μ is small...



The stability of the de-synchro state is nontrivial problem.



Generalized Spectral Theory
(H.C, 2015)

A theory to “erase” the continuous spectrum by the Gelfand triplet. Instead, *generalized eigenvalues* appears.

The generalized spectral theory (overview).

Let us consider a linear op. T defined on a Hilbert sp. $\mathcal{H}$

The resolvent $(\lambda - T)^{-1}$ diverges on the spectral set as an operator on $\mathcal{H}$.

STEP 1:

Find a subspace $X \subset \mathcal{H}$ s.t. if the resolvent is regarded as an operator from X into the dual space X' , then it has an analytic continuation beyond the continuous spectrum.

The continuous spectrum becomes a branch cut of the nontrivial Riemann surface of the resolvent.

The triplet

$$X \subset \mathcal{H} \subset X'.$$

is called the Gelfand triplet.

e.g.

Space of Schwartz distribution.

$$C_0^\infty(\mathbf{R}) \subset L^2(\mathbf{R}) \subset C_0^\infty(\mathbf{R})'.$$

The generalized spectral theory (overview).

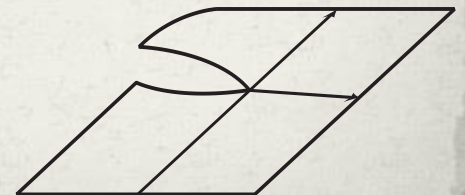
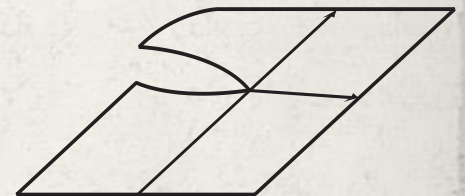
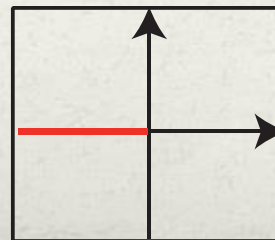
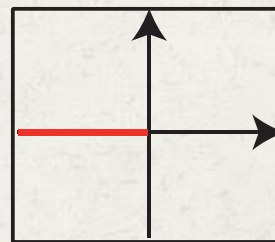
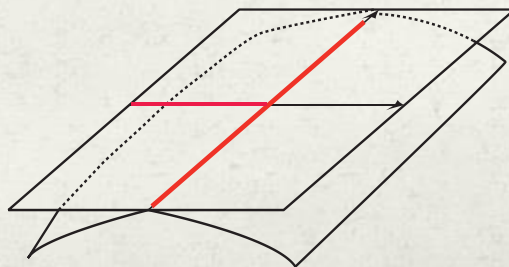
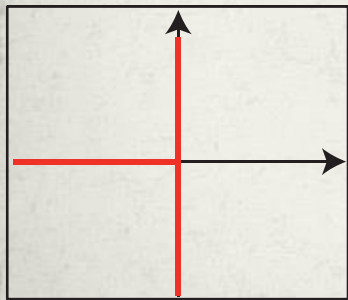
STEP 2:

Define a generalized eigenvalue as a singularity of the analytic continuation of the resolvent on the Riemann surface.

A corresponding eigenvector is not in $\mathcal{H}$ but in X' .

Generalized eigenvalues play a similar role to usual eigenvalues.
(H.C, Adv. in Math. 2015)

For our problem, the Riemann surface is disjoint two logarithmic surfaces.



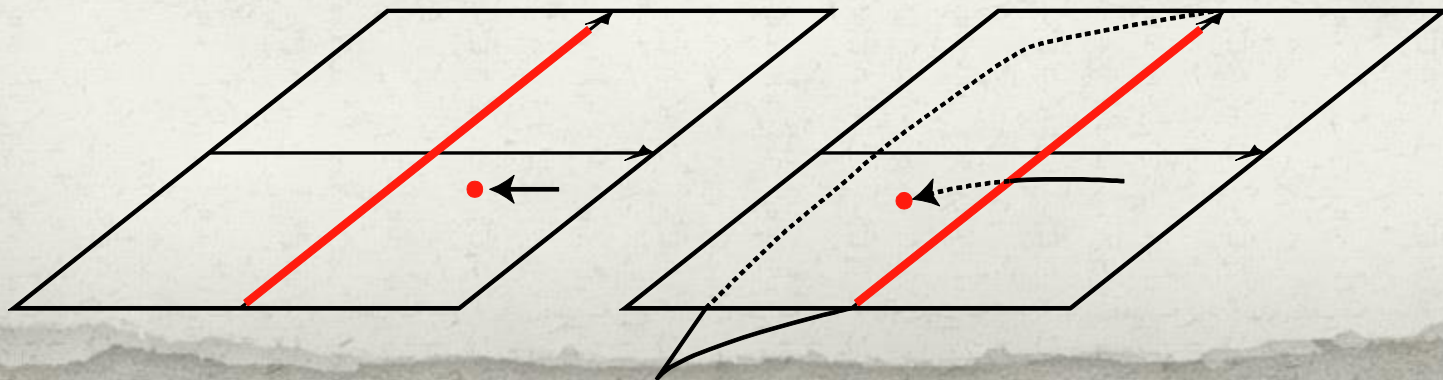
The generalized spectral theory (overview).

Generalized eigenvalues of our operator are given by roots of

$$\lambda + \frac{1}{\tau} = \begin{cases} \frac{-\mu}{4\pi g_L c_1} (\lambda + g_0 - c_2 g_L) \int \frac{-i\omega}{\lambda - i\omega} \tilde{G}(\omega) d\omega & (\text{Re}(\lambda) > 0) \\ \frac{-\mu}{4\pi g_L c_1} (\lambda + g_0 - c_2 g_L) \left(\int \frac{-i\omega}{\lambda - i\omega} \tilde{G}(\omega) d\omega - 2\pi \lambda \tilde{G}(-i\lambda) \right) & (\text{Re}(\lambda) < 0) \end{cases}$$

The second line is an analytic continuation of the first one from the right half plane to the left.

A root of the second line is not a usual eigenvalue but a generalized eigenvalue.



Main Result.

Suppose that $G(I)$ is symmetric and unimodal around the average $\eta > 0$, and meromorphic on $\mathbb{C}$.

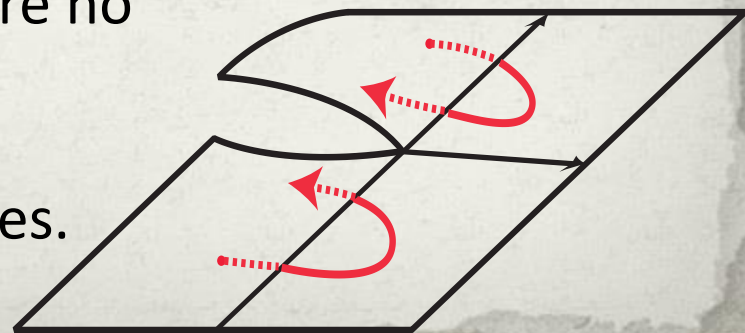
(i) When $\mu = 0$, there are generalized eigenvalues on the left half plane, whose position is described by a pole of $G(I)$.

There are no (generalized) eigenvalues on the right half plane.

(ii) Suppose $\eta > g_L/2c_1$ and $\tau > -1/c_2g_L$. If the variance of $G(I)$ is sufficiently small, generalized eigenvalues transversely get across the imaginary axis (and become usual eigenvalues) as μ increases.

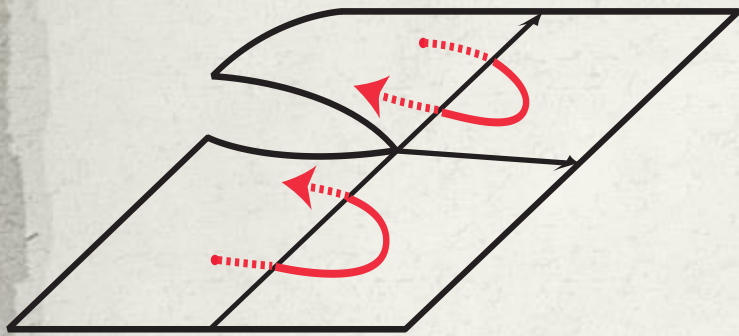
(iii) When μ is sufficiently large, there are no eigenvalues on the right half plane.

Hopf bifurcation occurs at least two times.

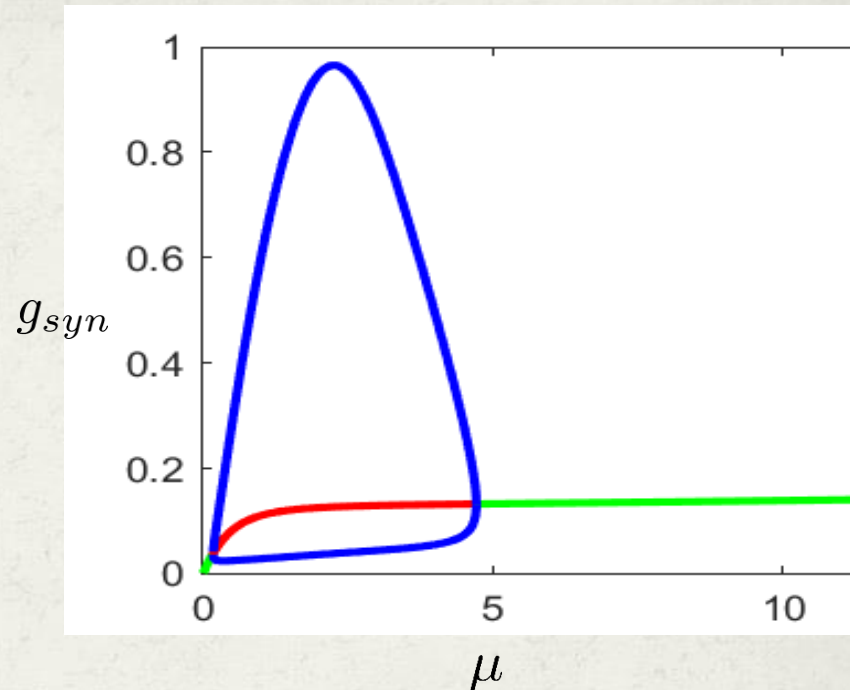


Numerical result for $G(I) = \frac{1}{\pi} \frac{\Delta}{(I - \eta)^2 + \Delta^2}$, $\eta = 2$, $\Delta = 0.1$.

[Akao, Kotani 2016]



The motion of the generalized eigenvalues determines (non)existence of the gamma oscillations.



Hopf bifurcations at $\mu \sim 0.18$ and $\mu \sim 4.7$

If a network is too complex, collective firing is violated.
(related to the synaptic pruning in early childhood ?)

(Part of the idea) Delta function.

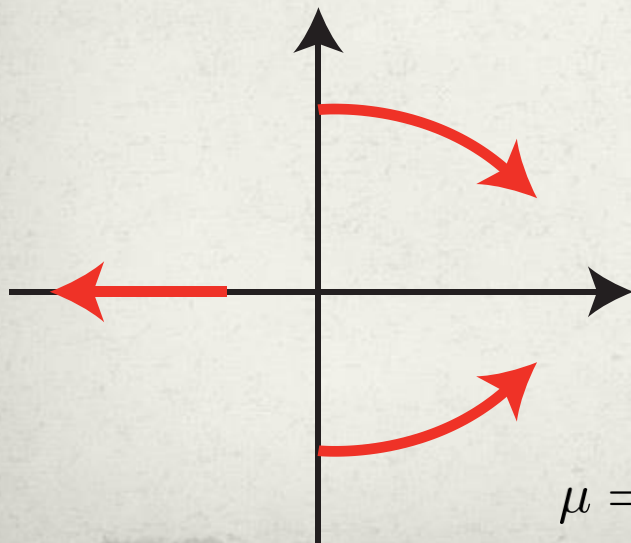
Suppose that $G(I)$ is a Dirac delta supported at $\eta > 0$.
The system is reduced to a 3-dim and it is easy to solve the eigen-equation given by

$$\lambda^3 + \frac{1}{\tau}\lambda^2 + (x + \frac{1}{\tau}g_0)\lambda + \frac{1}{\tau}(x + g_0^2 - c_2g_Lg_0) = 0, \quad x = \left(\frac{2\pi g_0}{\tau\mu}\right)^2$$

Prop.

Suppose $\eta > g_L/2c_1$ and $\tau > -1/c_2g_L$.

Then, the motion of three eigenvalues for $\mu > 0$ are given below.
A Hopf bifurcation occurs at $\mu = 0$.



Synchro (Gamma oscillation)
exists for any $\mu > 0$.

$\mu = 0 \rightarrow +\infty$

References

- The generalized spectral theory

H. Chiba,

A spectral theory of linear operators on rigged Hilbert spaces under certain analyticity conditions.

Adv. in Math. 273, 324-379, (2015).

H. Chiba,

A spectral theory of linear operators on rigged Hilbert spaces under certain analyticity conditions II.

Kyushu Journal of Math. (2018)

- A proof of the Kuramoto conjecture

H.Chiba,

A proof of the Kuramoto conjecture for a bifurcation structure of the infinite dimensional Kuramoto model,

Ergo. Theo. Dyn. Syst, (2013)

Applications of generalized spectral theory:

- K. Kotani, A. Akao, H.Chiba,
“ Bifurcation of the neuronal population dynamics of the modified theta model: transition to macroscopic gamma oscillation “
Physica D, Vol.416, 132789, (2021)

- H.Chiba, G. S. Medvedev, M. S. Muzuhara,
“Bifurcations in the Kuramoto model on graphs“
Chaos, 28, (2018)

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N a_{ij} \sin(\theta_j - \theta_i).$$

- On a second order Kuramoto model (submitted, with Medvedev)
- On generalized eigenvalues of a Perron-Frobenius operator (Koopman operator) of discrete dynamical systems (submitted, with Ishikawa, Ikeda).
Generalized eigenvalues describe the transient to a mixing state.

Thank you !

Idea of the proof
~ A generalized spectral theory ~

Infinite- dimensional Kuramoto model

$$(KM_N) \begin{cases} \dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) = \omega_i + Kr \sin(\psi - \theta_i), \\ re^{i\psi} := \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)}. \end{cases}$$



continuous limit

$$(KM_\infty) \begin{cases} \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial \theta} (v \rho) = 0, \\ v := \omega + Kr \sin(\psi - \theta), \\ re^{i\psi} = \int_{\mathbf{R}} g(\omega) d\omega \int_0^{2\pi} e^{i\theta} \rho(\theta, \omega, t) d\theta. \end{cases}$$

Evolution eq. for $\rho = \rho(\theta, \omega, t)$: prob. dens. func. of $\theta \in S^1$ for each $(\omega, t) \in \mathbf{R} \times \mathbf{R}_+$.

Fourier coefficients:

$$Z_k(t, \omega) := \int_0^{2\pi} e^{ik\theta} \rho(\theta, \omega, t) d\theta \in L^2(\mathbf{R}, g(\omega) d\omega).$$

Evolution equations on $\prod_{-\infty}^{\infty} L^2(\mathbf{R}, g(\omega) d\omega)$.

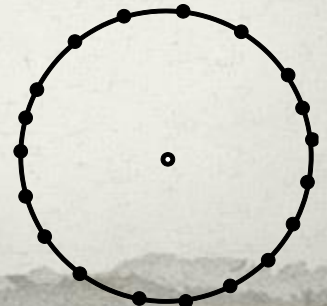
$$\begin{cases} \dot{Z}_1 = i\mathcal{M}Z_1 + \frac{K}{2}\mathcal{P}Z_1 + (\text{nonlinear}), \\ \dot{Z}_k = ik\mathcal{M}Z_k + (\text{nonlinear}), \quad (k = 2, 3, \dots), \\ re^{i\psi} := \mathcal{P}Z_1. \end{cases}$$

$\mathcal{M} : f(\omega) \mapsto \omega f(\omega)$, multiplication operator

$\mathcal{P} : f(\omega) \mapsto \int_{-\infty}^{\infty} f(\omega)g(\omega)d\omega$, projection

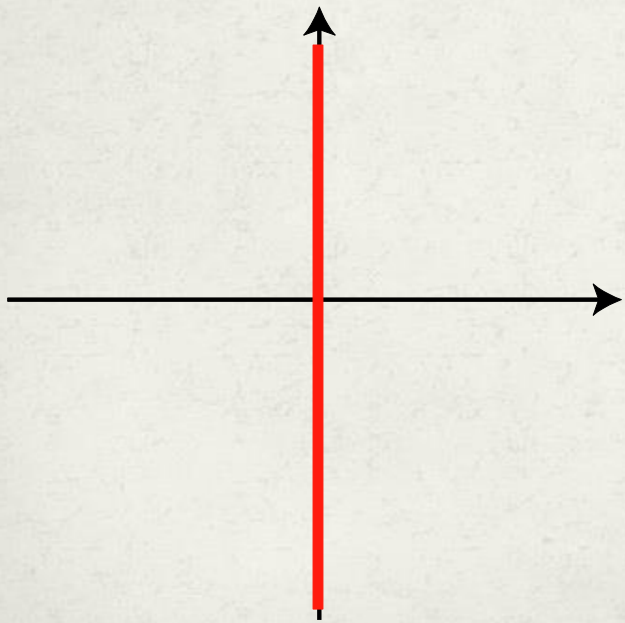
Trivial solution (de-synchro state)

$$Z_k \equiv 0 \iff \rho(\theta, \omega, t) = 1/(2\pi) \iff$$

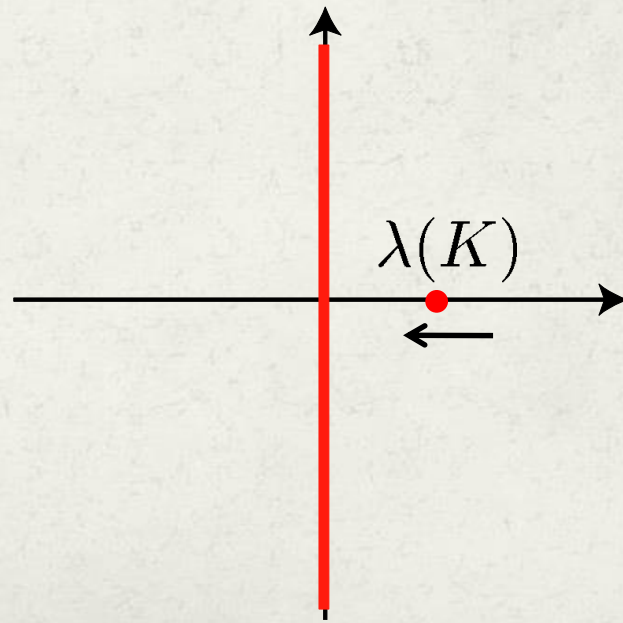


Spectrum of $T := i\mathcal{M} + K\mathcal{P}/2$

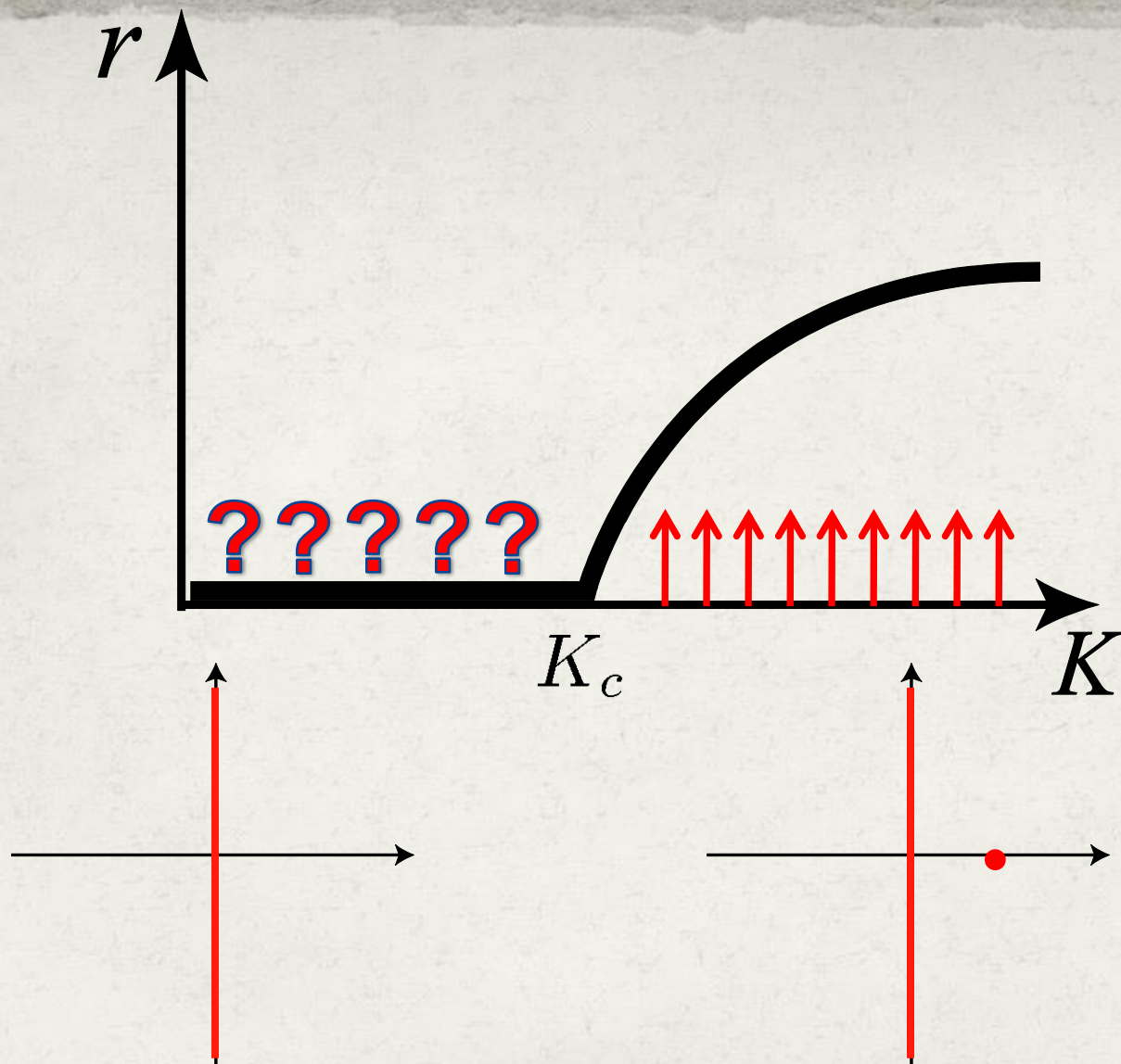
- conti. spec: $\sigma_c(T) = \sigma(i\mathcal{M}) = i\mathbf{R}$.
- eigenvalues: roots of $\int_{-\infty}^{\infty} \frac{1}{\lambda - i\omega} g(\omega) d\omega = 2/K$,
which exist iff $K_c := 2/(\pi g(0)) < K$.



$$0 < K < K_c$$



$$K_c < K$$



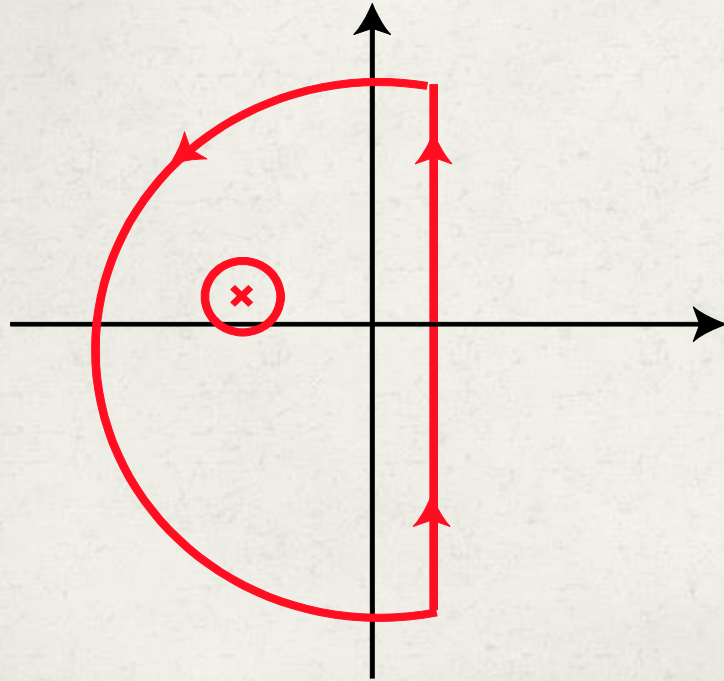
Stability? Bifurcations?

Let's reformulate in terms of the Gelfand triplet.

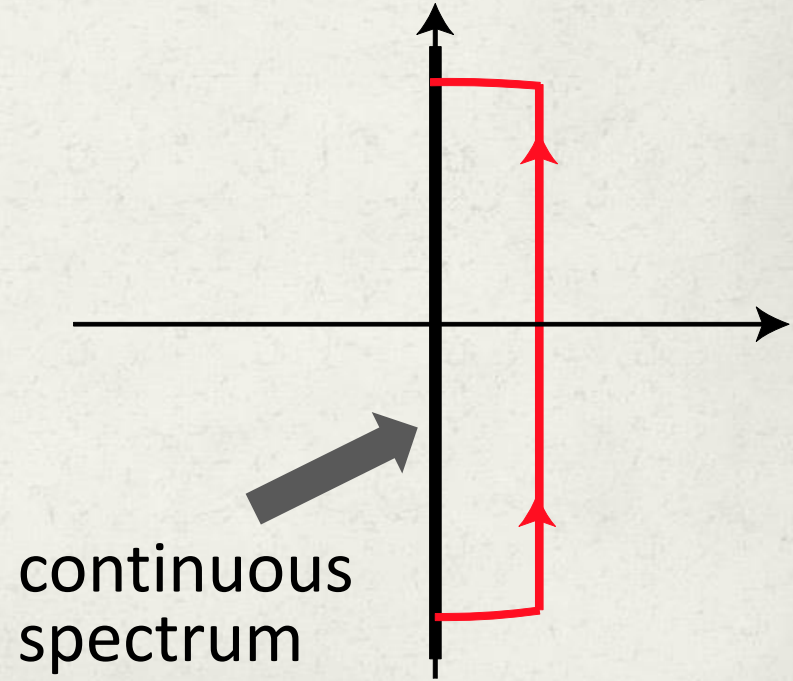
$$\dot{x} = \mathcal{A}x, \quad \mathcal{A} : X \rightarrow X \quad \text{Laplace inversion.}$$

$$e^{\mathcal{A}t}\phi = \frac{1}{2\pi i} \int_{x-i\infty}^{x+i\infty} e^{\lambda t} (\lambda - \mathcal{A})^{-1} \phi d\lambda$$

Resolvent. $R_\lambda \phi = (\lambda - \mathcal{A})^{-1} \phi$, analytic w.r.t. λ on $\mathbb{C} \setminus \sigma(\mathcal{A})$



$$e^{\mathcal{A}t}\phi = \sum (\text{residues})$$



$$e^{\mathcal{A}t}\phi = ?$$

Eq. $\dot{x} = \mathcal{A}x$, $\mathcal{A} : X \rightarrow X$

Sol. $e^{\mathcal{A}t}\phi = \frac{1}{2\pi i} \int_{x-i\infty}^{x+i\infty} e^{\lambda t} (\lambda - \mathcal{A})^{-1} \phi d\lambda$

Resolvent. $R_\lambda \phi = (\lambda - \mathcal{A})^{-1} \phi$, analytic w.r.t. λ on $\mathbf{C} \setminus \sigma(\mathcal{A})$

The resolvent diverges on $\lambda \in \sigma(\mathcal{A})$ as a function in X ...
But it may converge in a larger space.

e.g. Gaussian.

$$f(x) = \frac{1}{\sqrt{2\pi s}} \exp \left[-\frac{x^2}{2s^2} \right] \xrightarrow{s \rightarrow +0} \begin{cases} \infty & \text{(as a usual func.)} \\ \delta & \text{(Dirac delta.)} \end{cases}$$

Idea: Consider the multiplication operator on $L^2(\mathbf{R})$.

$$x : \phi(x) \mapsto x\phi(x).$$

Resolvent. $R_\lambda \phi = (\lambda - x)^{-1} \phi(x)$.

Spectrum. $\sigma(\mathcal{A}) = \mathbf{R}$.

When $\lambda \in \mathbf{R}$, $R_\lambda \phi \notin L^2(\mathbf{R})$.

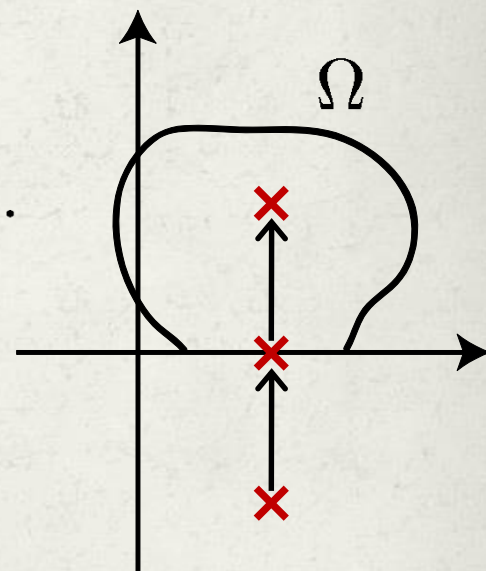
However, the inner product may exist!

$$(R_\lambda \phi, \psi) = \int_{\mathbf{R}} \frac{1}{\lambda - x} \phi(x) \psi(x) dx, \quad (\text{Im}(\lambda) < 0).$$

$$\downarrow \text{ (if } \phi \text{ and } \psi \text{ are continuous)}$$
$$\lim_{\text{Im}(\lambda) \rightarrow -0} \int_{\mathbf{R}} \frac{1}{\lambda - x} \phi(x) \psi(x) dx$$

$$\downarrow \text{ (if } \phi \text{ and } \psi \text{ are holomorphic)}$$

$$\int_{\mathbf{R}} \frac{1}{\lambda - x} \phi(x) \psi(x) dx + 2\pi i \phi(\lambda) \psi(\lambda), \quad (\text{Im}(\lambda) > 0).$$



The analytic continuation

$$(R_\lambda \phi, \psi) = \int_{\mathbf{R}} \frac{1}{\lambda - x} \phi(x) \psi(x) dx + 2\pi i \phi(\lambda) \psi(\lambda), \quad (\text{Im}(\lambda) > 0).$$

exists only when ϕ and ψ are holomorphic.

$X \subset L^2(\mathbf{R})$: some class of holomorphic functions.

X' : the dual space (the set of conti. linear functionals on X)

The mapping $\psi \mapsto (R_\lambda \phi, \psi)$ defines a linear functional on X .

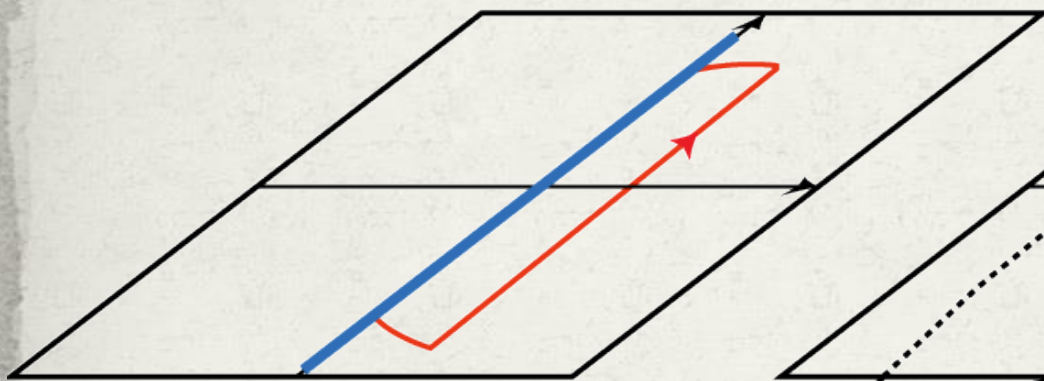
The mapping $\phi \mapsto (R_\lambda \phi, \cdot) \in X'$ defines a linear map from X into X' .

The resolvent $R_\lambda : L^2(\mathbf{R}) \rightarrow L^2(\mathbf{R})$ has an analytic continuation from the lower half plane to the upper half plane as an operator from X into X' .

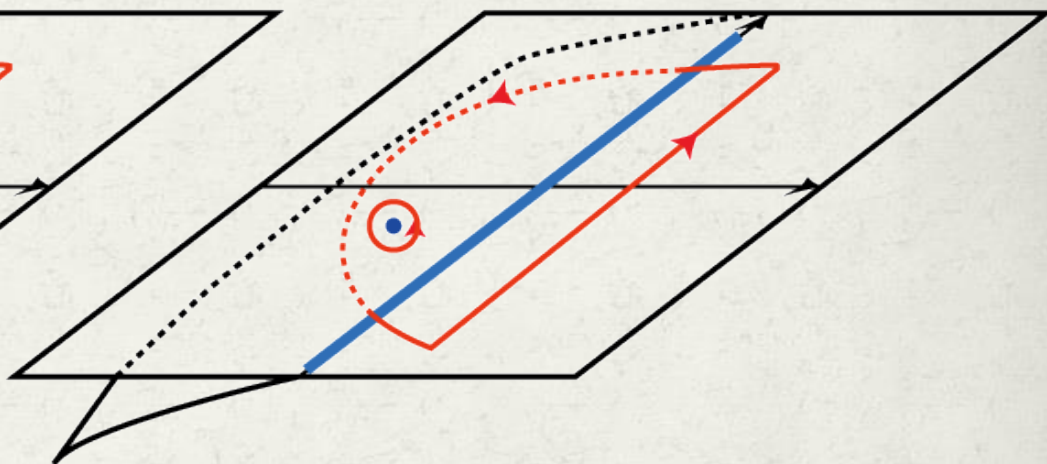
Gelfand triplet. $X \subset L^2(\mathbf{R}) \subset X'$

spectrum $\longleftrightarrow$ singularities of the resolvent (on $\mathbb{C}$).
generalized spectrum $\longleftrightarrow$ singularities of an analytic continuation of the resolvent (on a Riemann surface).

The first Riemann sheet



The second Riemann sheet



Through the Laplace inversion formula, the generalized spectrum induces an exponential decay of a solution.

- H. Chiba, **Adv. in Math.** 273, 324-379, (2015).

A spectral theory on a Gelfand triplet.

For the Kuramoto model,

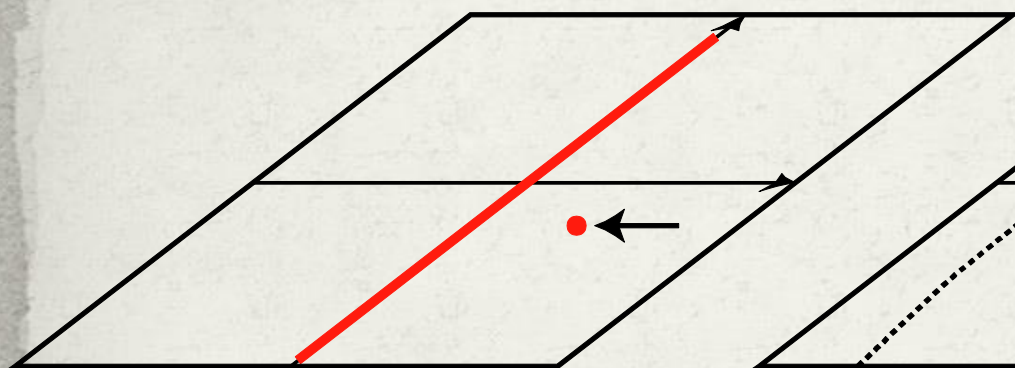
$$\left\{ \begin{array}{l} T = i\mathcal{M} + \frac{K}{2}\mathcal{P}. \\ X : \text{a certain space of holo. func.} \\ \text{Gelfand triplet : } X \subset L^2(\mathbf{R}, g(\omega)d\omega) \subset X' \\ \text{paring : } \langle \mu | \phi \rangle \\ g(\omega) : \text{Gaussian distribution.} \end{array} \right.$$

The resolvent $(\lambda - T)^{-1} : L^2 \rightarrow L^2$ has an analytic continuation $\mathcal{R}_\lambda : X \rightarrow X'$.

- generalized eigenvalues λ_n
- generalized eigenfunctions $\mu_n \in X'$
- generalized eigenspaces and projections.

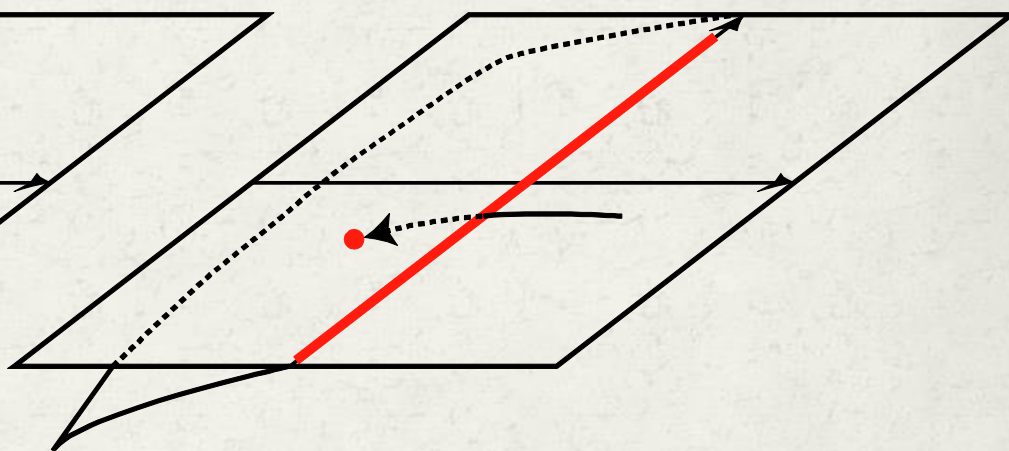
For the Kuramoto model, the number of generalized eigenvalues is countable.
They lie on the **second** Riemann sheet of the resolvent.

Spectrum in L^2 -sense



$$K_c < K$$

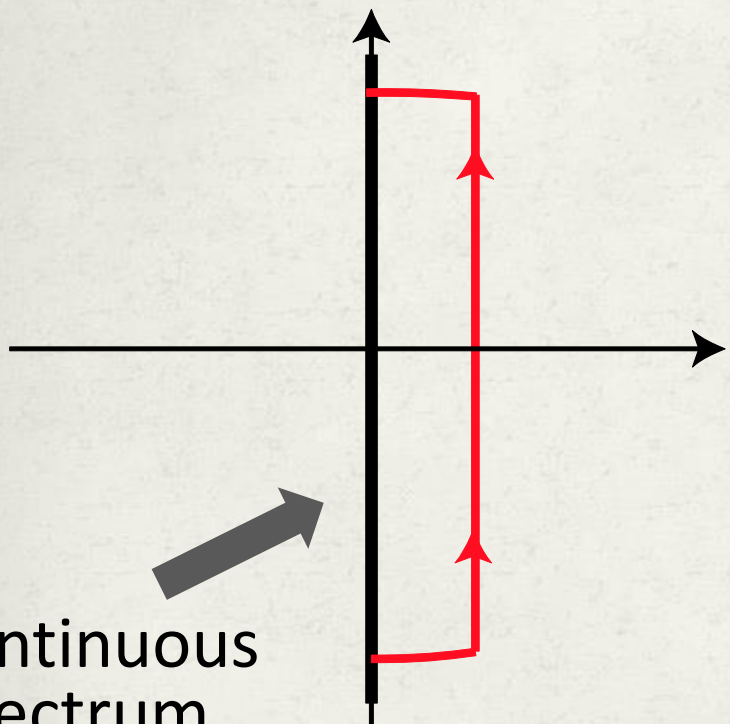
Generalized spectrum



$$0 < K < K_c$$

Semigroup in L^2 -sense.

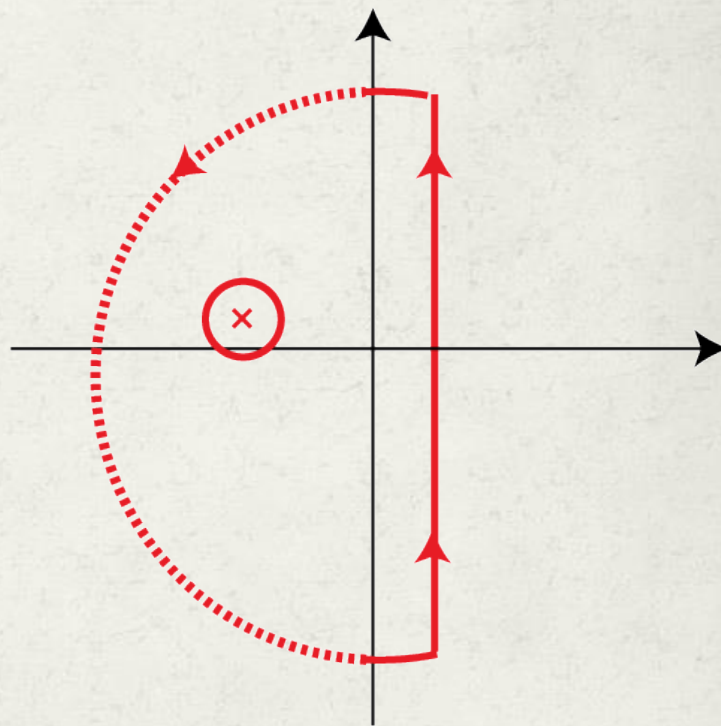
$$e^{Tt}\phi = \frac{1}{2\pi i} \int_{\gamma} e^{\lambda t} (\lambda - T)^{-1} \phi d\lambda$$



continuous
spectrum

Semigroup on the dual sp.

$$(e^{Tt})^{\times} \phi = \frac{1}{2\pi i} \int_{\gamma} e^{\lambda t} \cdot \mathcal{R}_{\lambda} \phi d\lambda$$



$$(e^{Tt})^{\times} \phi = \sum (\text{residues})$$

Theorem. (spectral decomposition)

For any $\phi \in X$,

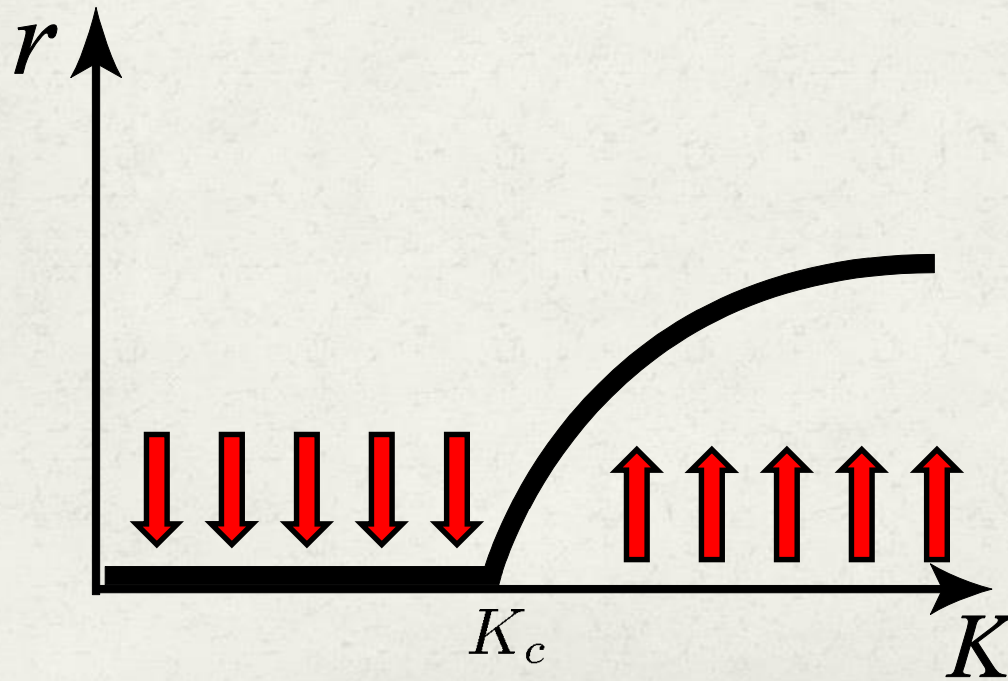
$$(e^{Tt})^\times \phi = \sum_{n=0}^{\infty} e^{\lambda_n t} \langle \mu_n | \phi \rangle \cdot \mu_n.$$

Theorem. (completeness)

- (i) A system of generalized eigenfunctions $\{\mu_k\}_{k=0}^{\infty}$ is complete
($\langle \phi | \mu_k \rangle = 0, \forall k \Rightarrow \phi = 0$).
- (ii) linearly independent : $\sum a_k \mu_k = 0 \Rightarrow a_k = 0$
- (iii) the decomposition is unique.

$$(e^{Tt})^\times \phi = \sum_{n=0}^{\infty} e^{\lambda_n t} \langle \mu_n | \phi \rangle \cdot \mu_n.$$

When $0 < K < K_c$, all gener. eigenvalues lie on the left half plane, which proves the stability of the de-synchro state.



The next purpose is a bifurcation at $K = K_c$.

On $L^2(\mathbf{R}, g(\omega)d\omega)$, the center subspace is infinite-dim because $\sigma_c(T) = i\mathbf{R}$.

Generalized center subspace:

$$\mathbf{E}_c = \text{span}\{ \mu_n \mid \lambda_n \in i\mathbf{R} \} \subset X'.$$



Existence of a finite-dim center mfd. on X' .

For the Kuramoto model, this is one-dimensional.

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N.$$



$$N \rightarrow \infty$$

(continuous model)



$$Z_k(t, \omega) := \int_0^{2\pi} e^{ik\theta} \rho(\theta, \omega, t) d\theta.$$

$$\frac{dZ_1}{dt} = TZ_1 - \frac{K}{2}(P_0, Z_1)Z_2.$$



$$X \subset L^2(\mathbf{R}, g(\omega)d\omega) \subset X' \quad \text{inclusion.}$$

$$\frac{d}{dt}|Z_1\rangle = T^\times|Z_1\rangle - \frac{K}{2}\langle P_0 | Z_1 \rangle |Z_2\rangle.$$



$$|Z_1\rangle = \alpha(t)|\mu_0\rangle + |Y_1\rangle.$$

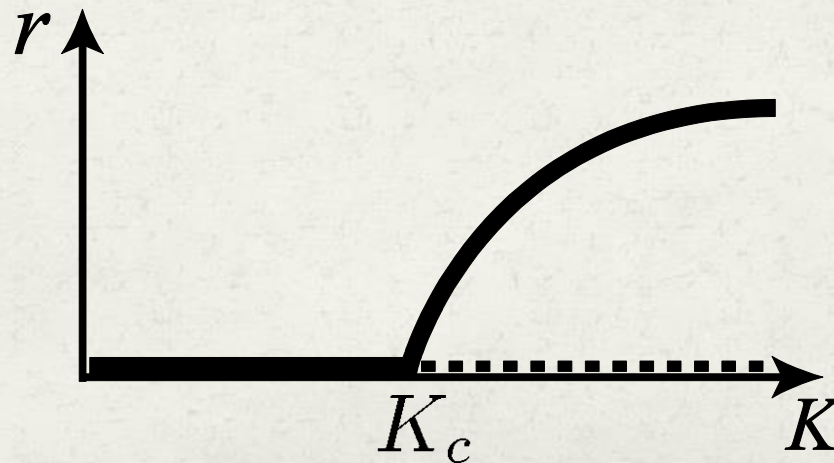
$$\frac{d}{dt}\alpha = (K - K_c)p_1\alpha + p_3\alpha|\alpha|^2 + O(|\alpha|^5).$$

Dynamics on a center manifold.

$$\frac{d}{dt}\alpha = (K - K_c)p_1\alpha + p_3\alpha|\alpha|^2 + O(|\alpha|^5).$$

$$|\alpha| = \sqrt{\frac{-p_1}{p_3}} \sqrt{K - K_c}.$$

$$p_1 = D_0/K_c, \quad p_3 = \pi D_0 K_c^3 g''(0)/16.$$



Kuramoto conjecture is proved.

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N (\sin(\theta_j - \theta_i) + h \sin 2(\theta_j - \theta_i)).$$

 $N \rightarrow \infty$

(continuous model)

 $Z_k(t, \omega) := \int_0^{2\pi} e^{ik\theta} \rho(\theta, \omega, t) d\theta.$

$$\frac{dZ_1}{dt} = TZ_1 - \frac{K}{2}(P_0, Z_1)Z_2 - \frac{Kh}{2}(Z_2, P_0)Z_{-1}$$

 $i : X \rightarrow X'$

$$\frac{d}{dt}|\bar{Z}_1\rangle = T^\times|Z_1\rangle - \frac{K}{2}\langle P_0 | Z_1\rangle|Z_2\rangle - \frac{Kh}{2}\langle P_0 | Z_2\rangle|Z_{-1}\rangle.$$

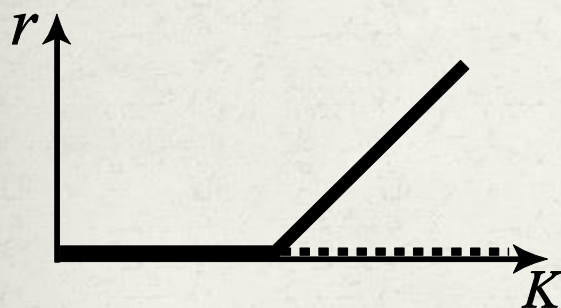
 $|Z_1\rangle = \alpha(t)|\mu_0\rangle + |Y_1\rangle.$

$$\frac{d}{dt}\alpha = (K - K_c)p_1\alpha + p_3\alpha|\alpha|^2 + p_2\alpha^2 e^{-i\arg(\alpha)} + O(|\alpha|^4).$$

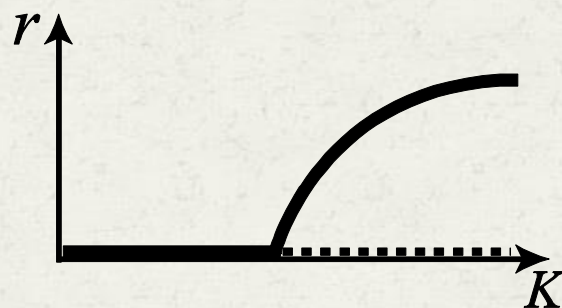
Dynamics on a center manifold.

$$\frac{d}{dt}\alpha = (K - K_c)p_1\alpha + p_2\alpha^2 e^{-i\arg(\alpha)} + O(|\alpha|^3).$$

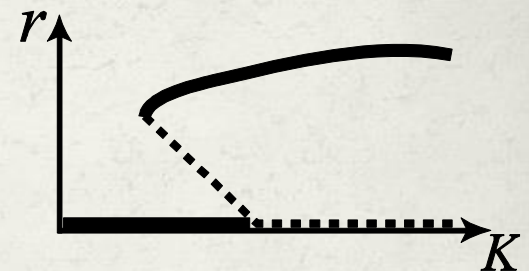
$$|\alpha| = \frac{2(1-h)}{K_c^3 \gamma h} (K - K_c).$$



$$h < 0$$



$$h = 0$$



$$0 < h < 1$$

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